PCP

LINUX

Prepared by

Prof. Chauhan Bhavik | Assistant professor | MScIT

- ➢ 308-A, SSASIT
- ➢ Communication: bhavik.chauhan@ssasit.ac.in
- ➢ Linkedin: https://www.linkedin.com/in/bhavik-chauhan-4972a8aa/
- ➢ Github: https://github.com/ChauhanBhavik5

Content

History of Linux (10 min)

- Unix (1969): Developed at Bell Labs (Ken Thompson & Dennis Ritchie).
- Problems with Unix: Expensive, closed-source, limited to big companies.
- Minix (1987): Small Unix-like OS for education (Andrew Tanenbaum).
- Linux (1991): Linus Torvalds created a free, open-source kernel.
- Open Source Movement: GNU Project + Linux kernel = GNU/Linux.
- Today: Used in servers, cloud, mobile (Android), supercomputers.

Timeline: Unix (1969) → Minix (1987) → Linux (1991) → Modern Linux Distros.

Features of Linux (15 min)

- **Multi-user: Many users can log in simultaneously.**
  - More than one person can log into the same system at once.
  - Each user has their own files, settings, and permissions.
  - Users cannot easily interfere with each other's data.
  - Example: A server can host websites for thousands of users at the same time.
- **Multitasking: Run multiple processes at the same time.**
  - Linux can run many programs/processes at the same time.
  - Background jobs (like downloads) keep running while you work.
  - System resources (CPU, memory) are shared efficiently.
  - Example: You can browse the web, edit a file, and play music together.
- **Portability: Runs on PCs, servers, mobiles, IoT devices.**
  - Linux can run on almost any hardware: PC, laptop, mobile, supercomputer, IoT device.
  - The same software can be compiled and used across different platforms.
  - Doesn't require very high-end hardware (can run on old PCs too).
  - Example: Android (phones) and Ubuntu (PCs) are both Linux-based.
- **Security: File permissions, user groups, firewalls.**
  - Strong file permissions (read, write, execute) control access.

- User accounts & groups provide isolation.
- Built-in security tools like firewalls (iptables, ufw).
- Much less targeted by viruses compared to Windows.

➢ **Open Source: Source code is freely available.**
- Source code of Linux is freely available to everyone.
- Anyone can study, modify, or improve it.
- A large global community maintains and supports it.
- Free to use → no licensing fees.

➢ **Stability & Performance: Uptime of years, widely used in servers.**
- Linux can run for months or years without crashing.
- Very efficient at handling heavy loads (databases, servers, cloud).
- Used in 90%+ of supercomputers and most web servers.
- Example: Companies like Google, Facebook, Amazon run on Linux servers.

Question: "Why do you think most web servers use Linux instead of Windows?"

Linux Architecture (20 min)

Layers of Linux OS:

Diagram

Hardware (Base Layer)

➢ Includes CPU, memory (RAM), hard disk, and I/O devices (keyboard, mouse, network card).

➢ Provides the raw resources needed by the operating system.

➢ Hardware does not interact with applications directly → it needs the kernel.

➢ Example: Your browser doesn't talk to RAM directly; the kernel manages it.

Kernel (Kernel Space):

➢ The core of Linux – acts as a bridge between hardware and user programs.

- ➢ Key Points:
  - Resource Manager: Manages CPU scheduling, memory allocation, file system, devices.
  - Monolithic Kernel:
    - All essential services (memory, file system, device drivers) run inside kernel mode.
    - Makes Linux powerful and fast.
  - System Calls: Provides an interface for programs to request kernel services.
  - Device Drivers: Enable hardware (keyboard, USB, printer) to work with Linux.
  - Example: When you open a file, your text editor uses system calls → kernel → hard disk.

User Space

- ➢ This is where users interact with Linux.

Components:

- ➢ Shell:
  - Command-line interpreter (bash, zsh, fish).
  - Converts user commands into kernel-understandable instructions.
  - Example: When you type ls, the shell sends a request to the kernel.
- ➢ Libraries:
  - Pre-written code that applications reuse (e.g., glibc).
  - Provides standard functions (open file, print output).
- ➢ Applications:
  - Programs for end users (text editors, browsers, servers).
  - Built on top of libraries + shell.

Linux vs Windows vs Unix (10 min)

Feature	Linux	Windows	Unix
Cost	Free & Open Source	Licensed (paid)	Expensive
Source Code	Open	Closed	Closed
Users	Multi-user	Multi-user (limited)	Multi-user
Security	Very strong	Vulnerable (more malware)	Strong
Portability	Runs everywhere	Limited	Limited
Usage	Servers, mobile, cloud, desktops	Desktops, gaming	Legacy servers
Example	Fedora, Red Hat, Debian, Android	Windows 10, 11 (desktops & laptops), Windows Server	AIX (IBM), HP-UX (HP), Solaris (Oracle)

Lecture 2: Linux File System & Directories

Linux File System (15 min)

Hierarchical Structure

- ➢ Linux uses a single-rooted tree structure.
- ➢ Everything starts at the root directory /.
- ➢ Directories branch out like a tree (like C:\ in Windows).
- ➢ Devices, files, and processes are all represented as files in Linux.

Example:

```
/
├── home/
│   └── user/
├── etc/
├── var/
└── bin/
```

Inode Concept

- ➔ Each file/directory has an **inode** (index node).
- ➔ Inode stores **metadata** (permissions, owner, size, timestamps, location on disk).
- ➔ Inode does **not store file name**, just file info.
- ➔ File name is mapped to inode via directory entry.
- ➔ Example command:
  - ◆ ls -i file.txt # shows inode number

Linux Standard Directories

Directory	**Purpose**
/	Root directory (starting point of everything)
/home	User home directories (/home/student)
/etc	System configuration files (/etc/passwd, /etc/hosts)
/var	Variable data like logs (/var/log), spool, mail
/bin	Essential user commands (ls, cp, mv)
/usr	User programs, libraries, and documentation
/tmp	Temporary files (cleared on reboot)
/dev	Device files (e.g., /dev/sda for hard disk, /dev/tty for terminal)
/proc	Virtual file system showing kernel & process info (/proc/cpuinfo, /proc/meminfo)

File Permissions Basics

- ➢ Each file has 3 types of permissions:
  - Read (r), Write (w), Execute (x)
- ➢ Assigned to 3 categories of users:
  - Owner, Group, Others
- ➢ Command
  - ☐ ls -l
- ➢ Example output:
  - ☐ -rwxr-xr-- 1 user group 1024 Sep 20 file.sh

  Breakdown:
  - rwx → Owner can read, write, execute
  - r-x → Group can read, execute
  - r-- → Others can only read

Hardware Requirements of Linux

Hardware Requirements

Different Linux distributions (distros) have different hardware requirements. Here's a basic guide:

Component	Minimum Requirements	Recommended for Smooth Performance
CPU	1 GHz Processor	2 GHz Dual-Core Processor or higher
RAM	512 MB to 1 GB	2 GB or more
Storage	5 GB of disk space minimum	20 GB or more
Graphics	Any basic graphics support	Higher-end graphics (optional for gaming / graphics - heavy tasks)
Network	Network interface for internet access	Ethernet or Wi-Fi